

dimensionality, and allows interactive exploration of projected data.

Hierarchical cluster explorer

Description: Visualises hierarchical clustering results as dendrograms, helping understand relationships between data points and identify cluster hierarchies.

Working: Data points start at the leaves and merge upwards based on similarity measures. Dendrogram branches represent these merges, with distances indicating cluster closeness.

Technical points: Supports various clustering algorithms (e.g., single linkage, complete linkage), allows interactive cutting of dendrograms to define clusters, and offers distance matrix visualisation.

PCA biplot

Description: Visualises both data points and principal components (PCs) in a low-dimensional space.

Working: PCA first reduces data dimensionality, capturing most variance. Biplot shows data points and corresponding projections onto the first few PCs.

Technical points: Allows interactive exploration of different PC combinations, colouring points based on class labels, and displaying contribution vectors for each PC.

Knowledge flow editor

Description: Creates visual workflows representing data processing and analysis pipelines.

Working: Users drag-and-drop WEKA operators (e.g., filters, classifiers) to build a sequence of data processing steps.

Technical points: Supports different workflow components (e.g., loaders, savers, evaluators), allows branching and looping, and exports workflows as files.

Visualise filter output

Description: Displays the transformed data after applying filters like normalisation or discretisation.

Working: Users select a filter and data, and the tool shows the processed data based on the chosen filter's settings.

Technical points: Supports various filters (e.g., Normalise, Discretise), offers pre/post visualisation comparison, and displays descriptive statistics for transformed data.

Class visualisation

Description: Presents different class distributions, useful for exploring imbalanced datasets and evaluating classification performance.

Working: The tool can display class distributions as histograms, bar charts, or scatter plots, highlighting class imbalance and potential biases.

Technical points: Allows selection of specific attributes for visualisation,

Table 1: Comparing WEKA with other data mining tools

Feature	WEKA	TensorFlow	PyTorch	Scikit-learn
User-friendly interface	User-friendly GUI for easy navigation	Primarily code-based	Primarily code-based	User-friendly, but less visual
Diverse algorithm support	Comprehensive set of algorithms	Extensive support for various models	Popular for deep learning applications	Wide range, but focused on traditional ML
Generative adversarial networks (GANs)	Supports GANs implementation	Strong GAN support	Extensive GAN capabilities	Limited built-in GAN support
Text generation	Suitable for text mining tasks	NLP capabilities for text generation	Strong NLP capabilities	Basic text processing capabilities
Image generation	Supports image processing and analysis	Widely used for computer vision tasks	Extensive computer vision capabilities	Limited image processing capabilities
Music composition	Capable of analysing musical patterns	Not specialised for music generation	Not specialised for music generation	Not specialised for music generation
Community support	Active community support and resources	Large community with extensive resources	Growing community with strong support	Well-established community
Flexibility and adaptability	Flexible for various machine learning tasks	Highly flexible for diverse applications	Easily adaptable for custom solutions	Flexible but more focused on traditional ML